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B

For astronaut to see the light sources,

$$P_{\text{received}} = \frac{P_{\text{source}}}{4\pi r^2} \times A_{\text{pupil}} \geq P_{\text{min}}$$

$$r \leq \sqrt{\frac{P_{\text{source}} A_{\text{pupil}}}{4\pi P_{\text{min}}}} = \sqrt{\frac{10\pi (0.0050/2)^2}{4\pi (2.0 \times 10^{-13})}} = 8838.8 \text{ m}$$

$$r_{\text{max}} = 8800 \text{ m}$$

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A

Diffraction is pronounced when the wavelength of the wave is comparable to the width of